

Autonomous Field Navigation with Radar

[ID 022026]

Develop an autonomous navigation stack for a small field robot with obstacle detection supported by radar and positioning sensors.

Objectives

- Enable autonomous field navigation
- Detect obstacles using radar data
- Fuse localization and perception signals for safer motion

Tasks

- Integrate lidar GPS pixhawk and radar interfaces
- Build localization and navigation modules
- Detect obstacles from radar observations
- Plan and execute autonomous paths in the field
- Validate navigation performance on the robot chassis

Requirements

- Python programming
- Robotics fundamentals
- ROS basics

Equipment

- RPLIDAR C1 LiDAR
- ZED-F9P-01B-01 GPS module
- Holybro Pixhawk 6X flight controller
- UGV Rover ROS 2 Open-source 6 Wheels 4WD

Deliverables

- Autonomous navigation prototype with radar-based obstacle detection
- Field validation results

Work Breakdown

- 30% hardware integration
 - 70% coding
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Duration: 2 months

Payment: Available in Bizquick

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